FINITENESS OF TANGENT AND INFINITESIMAL AUTOMORPHISM SPACES FOR THE STACK OF PROPER FLAT SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file quot.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Section artin-section-tangent-spaces (canonical tag 07WY). cited at statement line 3814 and proof line 3823. Tangent space framework.
- Section artin-section-inf (canonical tag 07Y6). cited at statement line 3817. Inf framework.
- Lemma lemma-spaces-RS-star (canonical tag 0D1J). cited at proof line 3825. (RS*) holds.
- Lemma artin-lemma-properties-lift-RS-star (canonical tag 07Y9). cited at proof lines 3826-3827. Produce the spaces.
- Remark artin-remark-compare-deformation-spaces (canonical tag 0D18). cited at proof line 3831. Agreement of definitions.
- Lemma defos-lemma-deform-spaces (canonical tag 0D17). cited at proof line 3836. Ext formulas.
- Lemma spaces-more-morphisms-lemma-netherlander-fp (canonical tag 0D0Z). cited at proof lines 3844-3845. NL structure.
- Lemma spaces-perfect-lemma-ext-finite (canonical tag 0D0T). cited at proof line 3849. Finite dimensionality.

Source reference. Original section 07WY: sources/stacks-chow/tags.txt, source lines 10058–10058. Cited at statement line 3814 and proof line 3823: the framework.

Source reference. Original section 07Y6: sources/stacks-chow/tags.txt, source lines 10101–10101. Cited at statement line 3817: the framework.

Source reference. Original lemma 0D1J: sources/stacks-chow/tags.txt, source lines 16349–16349. Cited at proof line 3825: (RS*) holds.

Source reference. Original lemma 07Y9: sources/stacks-chow/tags.txt, source lines 10104–10104. Cited at proof lines 3826-3827: produce T_x(k), Inf_x(k).

Source reference. Original remark 0D18: sources/stacks-chow/tags.txt, source lines 16338–16338. Cited at proof line 3831: agreement in the finite type case.

Date: stacks-project:quot.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0D17: sources/stacks-chow/tags.txt, source lines 16337–16337. Cited at proof line 3836: Ext formula for T_x and Inf_x.

Source reference. Original lemma 0D0Z: sources/stacks-chow/tags.txt, source lines 16329–16329. Cited at proof lines 3844-3845: NL_{X/k} coherent, degrees 0 and -1.

Source reference. Original lemma 0D0T: sources/stacks-chow/tags.txt, source lines 16323–16323. Cited at proof line 3849: finite dimensionality.

Lemma 0.2. *Let k be a field and let $x = (X \to \mathrm{Spec}(k))$ be an object of $\mathcal{X} = \mathcal{S}paces'_{fp,flat,proper}$ over $\mathrm{Spec}(k)$.*

1. *If k is of finite type over $\mathbf{Z}$, then the vector spaces $T\mathcal{F}_{\mathcal{X},k,x}$ and $Inf(\mathcal{F}_{\mathcal{X},k,x})$ (see Artin's Axioms, Section 07WY) are finite dimensional, and*
2. *in general the vector spaces $T_x(k)$ and $Inf_x(k)$ (see Artin's Axioms, Section 07Y6) are finite dimensional.*

Proof. The discussion in Artin's Axioms, Section 07WY only applies to fields of finite type over the base scheme $\mathrm{Spec}(\mathbf{Z})$. Our stack satisfies (RS*) by Lemma 0D1J and we may apply Artin's Axioms, Lemma 07Y9 to get the vector spaces $T_x(k)$ and $\mathrm{Inf}_x(k)$ mentioned in (2). Moreover, in the finite type case these spaces agree with the ones mentioned in (1) by Artin's Axioms, Remark 0D18. With this out of the way we can start the proof. Observe that the first order thickening $\mathrm{Spec}(k) \to \mathrm{Spec}(k[\epsilon]) = \mathrm{Spec}(k[k])$ has conormal module k. Hence the formula in Deformation Theory, Lemma 0D17 describing infinitesimal deformations of X and infinitesimal automorphisms of X become

$$T_x(k) = \mathrm{Ext}^1_{\mathcal{O}_X}(NL_{X/k}, \mathcal{O}_X) \quad \text{and} \quad \mathrm{Inf}_x(k) = \mathrm{Ext}^0_{\mathcal{O}_X}(NL_{X/k}, \mathcal{O}_X)$$

By More on Morphisms of Spaces, Lemma 0D0Z and the fact that X is Noetherian, we see that $NL_{X/k}$ has coherent cohomology sheaves zero except in degrees 0 and -1. By Derived Categories of Spaces, Lemma 0D0T the displayed Ext-groups are finite k-vector spaces and the proof is complete. □